Foundations of Programming Lab Assignments 16 – 22

A#16 – Even/Odd | A#17 – Sine Series | A#18 – Strings | A#19 – Employee | A#20 – Point

ASSIGNMENT NO. 17

Sum of Sine Series (Taylor Expansion)

sin(x) = x − x^3/3! + x^5/5! − x^7/7! + ... Compute up to N terms for user-given x (radians).

C CODE

```c
#include <stdio.h>
#include <math.h>

int main() {
    double x, sum = 0.0, term;
    int n, i, j, power;
    long long fact;

    printf("Enter value of x (in radians): ");
    scanf("%lf", &x);
    printf("Enter number of terms: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        power = 2 * i + 1;   // 1, 3, 5, 7 ...
        fact  = 1;
        for (j = 1; j <= power; j++)
            fact *= j;
        term = pow(x, power) / (double)fact;
        if (i % 2 == 0) sum += term;
        else            sum -= term;
    }

    printf("Sum of sine series = %lf\n", sum);
    return 0;
}
/* Compile: gcc a17.c -o a17 -lm */
```

SAMPLE I/O

```
Input:
Enter x (in radians): 1
Enter number of terms: 5

Output:
Sum of sine series = 0.841471
```

FORMULA

TAYLOR SERIES

$\sin(x) = x - x^3/3! + x^5/5! - \ldots$

I-TH TERM

$(-1)^i \times x^{(2i+1)} / (2i+1)!$

COMPILE

```
gcc a17.c -o a17 -lm
```

FLOWCHART

Syntax error in text
mermaid version 10.9.1